Introduction

For this project, we are performing a finite element analysis study on a common hand tool: a tire iron or lug wrench. One end functions as a socket wrench for lug nuts, while the other end can be used as a drill bit or pry bar in some situations. This is a part most people are familiar with, but it actually experiences significant stresses in real use, especially under high torque or prying loads. The goal of this project is to model realistic loading conditions and understand how the geometry and material respond under those loads using FEA. The main metric we will be analyzing is the stress in the tire iron, and when the tool fails. The bent geometry is especially important because it introduces stress concentrations that are well-suited for finite element analysis.

Our motivation for choosing this part comes partly from personal interest in automotive work, which helps us intuitively understand how the tool is used in real life. From an analysis standpoint, this part offers a lot of flexibility. There are many potential dimensions to explore, and the loads can be applied in several realistic ways. The familiarity and popularity of the object makes our FEA model more meaningful and physically grounded.

Background

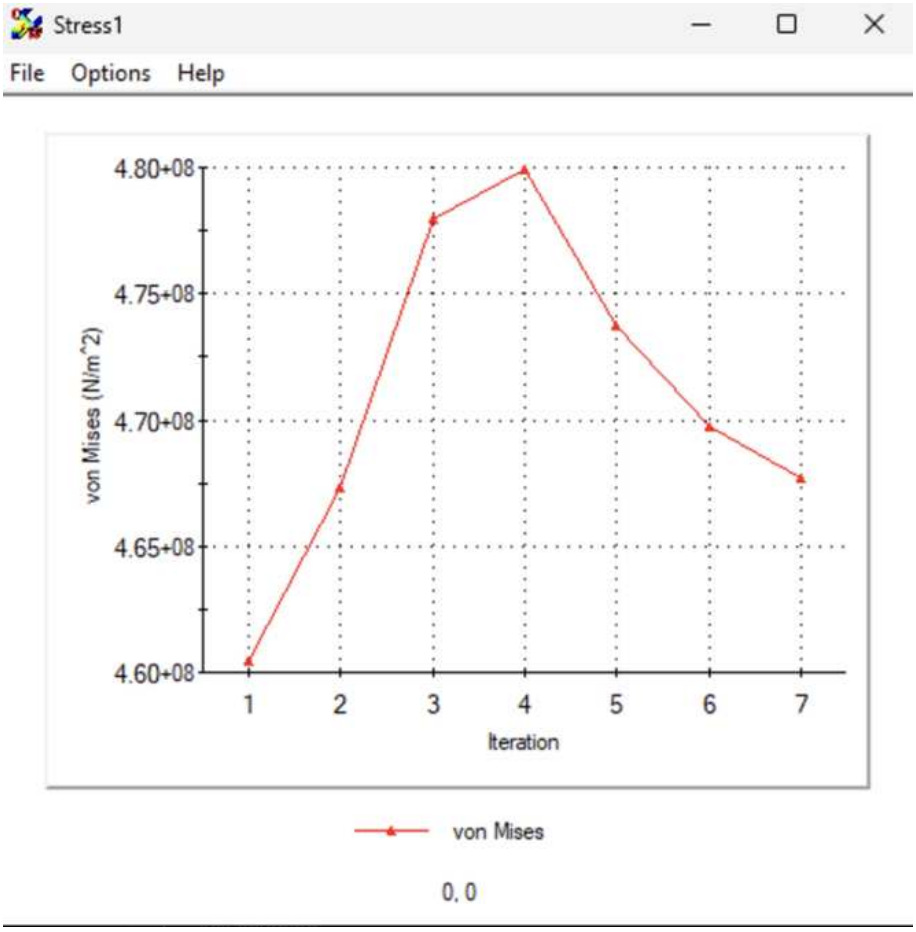
This is a very common tool, so obviously many companies have tested the properties of the tire iron under load. Any company making a tire iron tests it to ensure that it is functional and safe. Outside of companies that manufacture tire irons, there have been many second-party analytical and FEA studies of the tool. A study done on the use of a common wrench helped us quantify the magnitude and application of the force as well as the type of constraint (Azhagarsamy). Another study done on a L-shaped tire iron,

similar to our project, is an example of an analytical analysis of loading a tire iron (Sudiana). These studies were very useful in giving us starting references for our project.

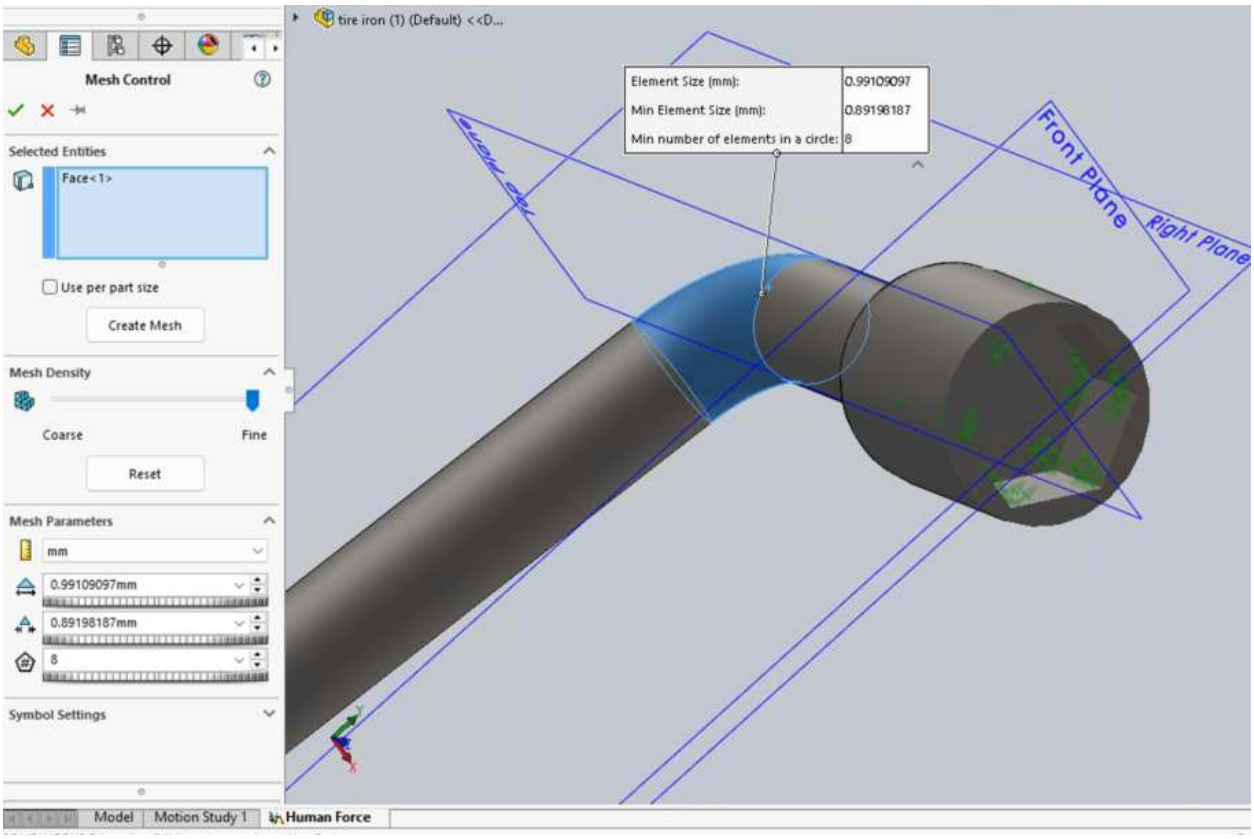
Method

To begin the FEA analysis of a tire iron, we needed to understand the dimensions, loading, and constraints. We found dimensions from part details for commercial tire irons. Then we modeled the part based on those dimensions in SolidWorks, creating a basic yet accurate 3D model. Based on the references described above, we chose to model the loading as a distributed load on the top side of the handle, near the end, over a distance of roughly one hand width. This is derived from the straightforward use of the tool, pushing down near the end of the handle with your hand. Again using the references, we chose to model the constraint as a fixed constraint on the inside faces of the socket. For the magnitude of the load, we determined the most force it would experience from a human user would be just under the weight of a user. This is based on the assumption that a user would step on the end of the bar to use their weight as maximum leverage on the lug nut. For the material, we chose a standard steel: AISI 1045, commonly used in many applications such as car tools.

Once we had a SolidWorks model and study set up, we performed the study with a basic, coarse mesh. We ran a convergence study through several iterations to find a more accurate value for maximum stress. The plot of stress vs iterations can be seen below:



As can be seen on this plot, the first few iterations show an increasing value for maximum stress. This was as a result of redesigning the mesh to be more fine, in hopes of getting a more accurate result. However after iteration 4, we examined the plot of the von Mises stress and noticed that the stress was mainly concentrated in the bend of the handle. So, we decided to apply a mesh control throughout this section to take a deeper look into the most integral part of the stress study, and hopefully generate more accuracy. This mesh control is shown below where we decreased the element sizes in this section, thus increasing the number of elements and the accuracy.



This then led to us seeing a convergence through iterations 5-7, whilst factoring in this new mesh control at increasingly finer mesh densities. This peak that we see is likely just the simulation having a high level of inaccuracy before we give it some direction in its analysis by focusing mesh density in important areas. Which eventually settled into convergence.

We also have recorded quantitative measures to measure the quality of our mesh, which can be seen below:

Study name	Human Force (-Default-)
DetailsMesh type	Solid Mesh
Mesher Used	Blended curvature-based mesh
Jacobian points for High quality mesh	16 points
Mesh Control	Defined
Max Element Size	3.96436 mm
Min Element Size	0.573954 mm
Mesh quality	High
Total nodes	50923
Total elements	33767
Maximum Aspect Ratio	3.7099
Percentage of elements with Aspect Ratio < 3	99.8
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Number of distorted elements	0
Reuse mesh for identical bodies	Off
Number of bodies that have reused mesh	0
Time to complete mesh(hh:mm:ss)	00:00:06
Computer name	RIWIN13

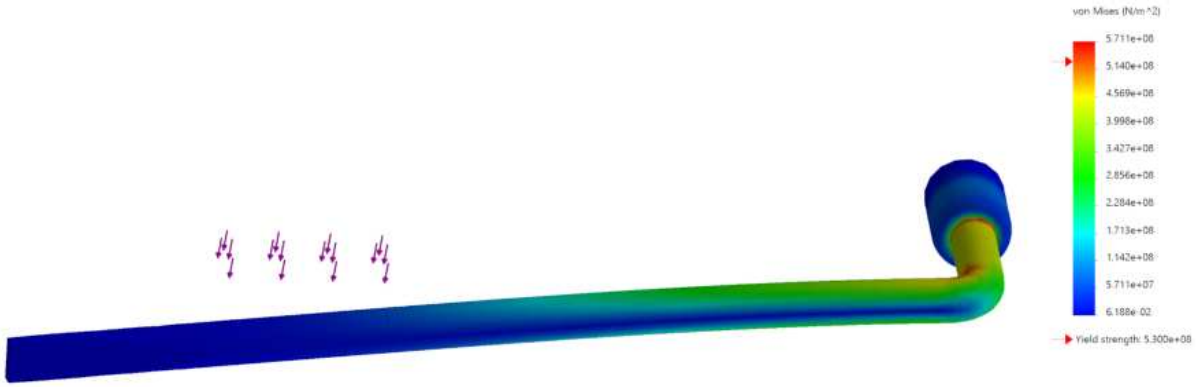
Important data points showing a quality mesh are as follows: Our percentage of elements with aspect ratio < 3 is greater than 99%; Our percentage of elements with aspect ratio > 10 is 0; Our maximum aspect ratio is 3.7099. These are all indicators of a high quality mesh.

Seeing the stress in the tire iron exceeded the yield strength of the material, we chose to redesign the part to prevent failure. The simple solution we came up with was to simply increase the diameter of the handle. This made the most sense logically, a thicker handle can undergo more stress and is more securely connected to the socket head. The same study with the same load and constraints was performed on the

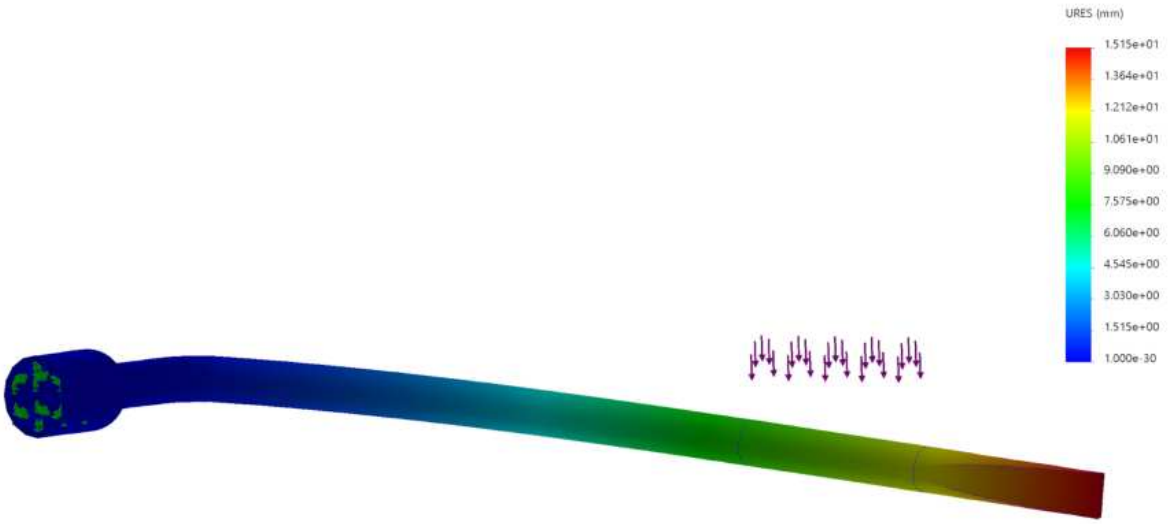
redesigned tire iron. Increasing the diameter of the handle only a few millimeters was enough to decrease the stresses in the tire iron below the yield strength.

Results

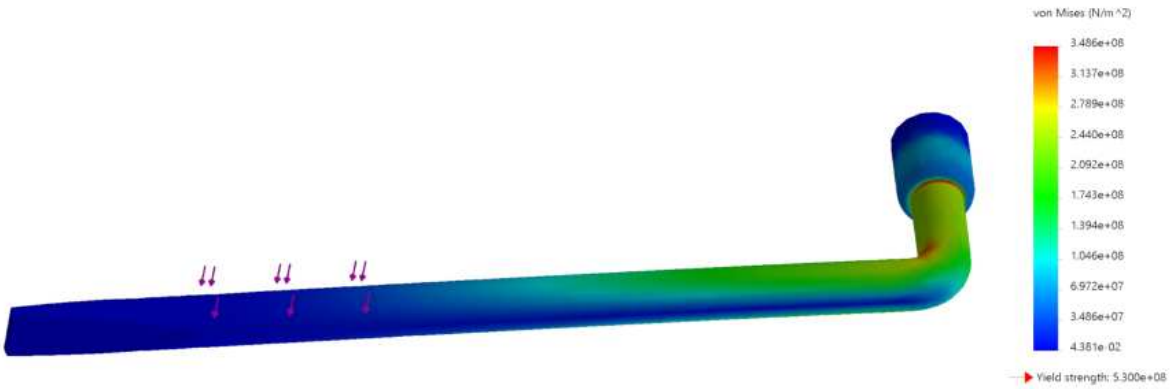
Here is the Von Mises Stress plot for the original tire iron. As you can see the stress goes beyond the yield strength, indicating the part has failed. We see the highest stresses in the bend of the handle, as well as where the handle connects to the socket head.



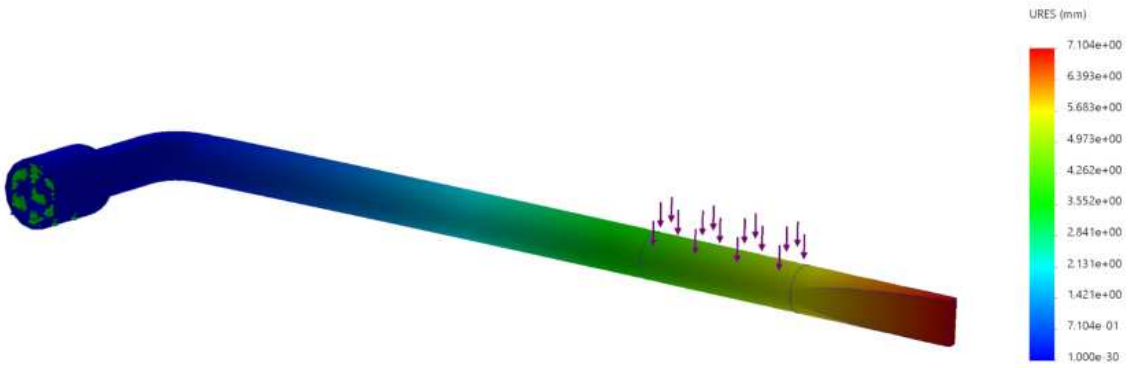
Here is the displacement plot for the original tire iron. As expected, the handle bends towards the end, with maximum displacement at the tip of the handle.



Here is the Von Mises Stress plot for the redesigned tire iron. The stresses in the part are now below the yield strength, indicating the part will not fail. The same areas experience high stress patches: the bend in the handle and where the handle meets the socket head.



The displacement plot for the redesigned part looks very similar to the original displacement, with slightly lower values.



Conclusion

Overall, the FEA results matched our expectations and intuitive logic. The results demonstrate the importance of a thick enough handle on tools experiencing high torques. The highest stresses occur at bends or interfaces between elements of the part such as the handle and the socket head interface. This is because at the bend the material experiences a high moment due to leverage, and the change in the force path due to the curvature increases stresses. The interface between the handle and socket head is where the applied torque is transferred into the lug nut. The high stress is due to the interface being a partially fixed boundary, where the inner faces of the socket are fixed but the interface is not. Other than these two high stress areas, the rest of the tool experiences relatively lower stresses.

The project went very smoothly for our first time doing a FEA study. Next time we would choose a more complex part, but for a first project the tire iron was a straightforward choice. One of the most challenging parts of the project was working on the mesh control, though even that went smoothly. Some future work that could build on this project could be optimizing the tool to minimize weight or cost while maintaining

functionality. Another direction the project could go could be simulating the use of the end that can be used as a screwdriver or crowbar. That is a different loading and constraint, so it would be interesting to see if the tire iron can hold up to high forces in that secondary use.

Citations

Azhagarsamy, Vignesh. (2020). Design and Finite element investigation of Solid Wrench.

Sudiana, Daffa. (2019). Stress - Strain Analysis on a L Lug Wrench.